

```
1  //LAB CODE 2000
2  //JASON JOHNSON
3  //LAB 2
4  #include <iostream>
5  #include <iomanip>
6
7
8  using namespace std;
9
10 int const ROWS{ 2 };
11 int const COLS{ 5 };
12
13 void getNumbers(int[][COLS]);
14 void showNumbers(int[][COLS]);
15 void addRows(int[][COLS]);
16 void highest(int[][COLS]);
17 void lowest(int[][COLS]);
18
19
20
21
22 int main() {
23
24     int numArr[ ROWS ][COLS] {};
25
26     getNumbers(numArr);
27
28     showNumbers(numArr);
29
30     addRows(numArr);
31
32     highest(numArr);
33
34     lowest(numArr);
35
36
37
38
39     return 0;
40 }
41
42
43
44 void getNumbers(int numArr[][COLS]) {
45     cout << "Please enter numbers into the array:\n";
46
47     for (int i = 0; i < ROWS; i++) {
48
49         for (int j = 0; j < COLS; j++) {
```

```
50
51         cout << "Enter number in position " << "(" << i << ", " << j  ➤
           << ") :";
52         cin >> numArr[i][j];
53
54     }
55 }
56 cout << endl;
57 }
58
59
60
61
62
63
64
65 void showNumbers(int numArr[][COLS]) {
66     cout << "Here are the numbers you entered:\n";
67
68     for (int i = 0; i < ROWS; i++) {
69
70         for (int j = 0; j < COLS; j++) {
71
72             cout << "\t" << numArr[i][j];
73
74         }
75     }
76     cout << endl;
77 }
78
79
80
81
82
83
84 void addRows(int numArr[][COLS]) {
85     int total{};
86
87     for (int i = 0; i < ROWS; i++) {
88         total = 0;
89         for (int j = 0; j < COLS; j++) {
90             total += numArr[i][j];
91
92         }
93         cout << "\n\nThe total for Row" << i + 1 << " is: " << total;
94
95     }
96
97 }
```

```
98 }
99
100
101
102
103 void highest(int numArr[][COLS]) {
104
105     int highest{numArr[0][0]};
106     for (int i = 0; i < ROWS; i++) {
107
108         for (int j = 0; j < COLS; j++) {
109
110             if (highest < numArr[i][j]) {
111                 highest = numArr[i][j];
112             }
113
114         }
115
116     }
117
118     cout << "\n\nThe highest number is: " << highest;
119 }
120
121
122
123
124
125
126 void lowest(int numArr[][COLS]) {
127     int lowest{numArr[0][0]};
128
129     for (int i = 0; i < ROWS; i++) {
130
131         for (int j = 0; j < COLS; j++) {
132
133             if (lowest > numArr[i][j]) {
134                 lowest = numArr[i][j];
135             }
136
137         }
138
139     }
140
141     cout << "\n\nThe lowest number is: " << lowest;
142 }
143
144
145
146 /*
```

```
147 OUTPUT:
148
149 Please enter numbers into the array:
150 Enter number in position (0, 0) :12
151 Enter number in position (0, 1) :45
152 Enter number in position (0, 2) :67
153 Enter number in position (0, 3) :54
154 Enter number in position (0, 4) :34
155 Enter number in position (1, 0) :56
156 Enter number in position (1, 1) :78
157 Enter number in position (1, 2) :98
158 Enter number in position (1, 3) :76
159 Enter number in position (1, 4) :43
160
161 Here are the numbers you entered:
162      12      45      67      54      34      56      78      98
           76      43
163
164
165 The total for Row1 is: 212
166
167 The total for Row2 is: 351
168
169 The highest number is: 98
170
171 The lowest number is: 12
172 C:\Users\jjohnson257\source\repos\LAB#2\x64\Debug\LAB#2.exe (process
      34108) exited with code 0 (0x0).
173 Press any key to close this window . . .
174
175
176 */
177
```